

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – ConstraintBased Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AIML/2024	Date:	18-08-2026

Code of Conduct

I, K.Mounith(1924252090)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Mounith

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability) □ Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

A.Constraint-Based Problem Solving

a) Referring Expressions and Antecedents

John —————→ He

Mary —————→ She

Ball —————→ it

John —————→ him

John —————┐

Mary —————┤————→ they

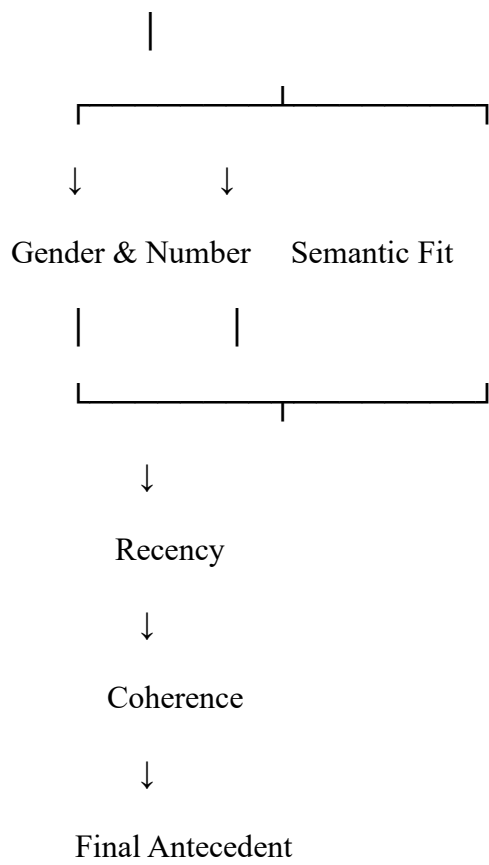
Dog —————┘

Final Antecedents:

- He → John
- She → Mary
- it → Ball
- him → John
- they → John + Mary + Dog

b) Applying Constraints

Referring Expression



He → John ✓ Gender

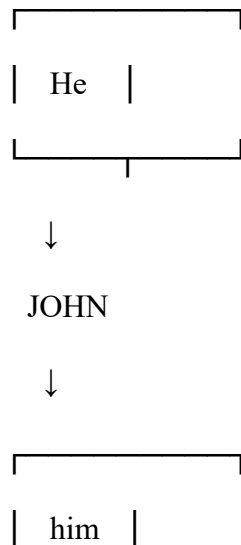
She → Mary ✓ Gender

it → Ball ✓ Semantic + Recency

him → John ✓ Gender + Semantic

they → John, Mary, Dog ✓ Number + Coherence

c) Constraint Graph



| She |

↓

MARY

| it |

↓

BALL

JOHN _____

MARY _____ → THEY

DOG _____

d) Final Coreference Chains

John → He → him

Mary → She

Ball → it

John + Mary + Dog → they

Resolved Paragraph:

John and Mary went to the park. **John** brought a ball. **Mary** wanted to play with the ball. The dog chased **John** excitedly. Finally, **John, Mary, and the dog** all went home.

e) Priority and Recency

Priority Order:

Gender/Number



Semantic Compatibility



Recency



Coherence

- **Gender and number** eliminate impossible antecedents.
- **Semantic compatibility** ensures the reference makes sense.
- **Recency** selects the nearest suitable antecedent.
- **Coherence** maintains consistent entity chains.

If Recency is Relaxed:

Normal:

Recent + Compatible → Best Antecedent

Without Recency:

Several Compatible Antecedents



Ambiguity

Therefore, relaxing recency can leave multiple possible antecedents and make coreference resolution more ambiguous.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).

3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate three different possible responses that satisfy all the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

A. a) Discourse Planning Steps

User Problem



Identify Intent



Advise + Encourage



Maintain Entities

(exam, concentrate, you)



Apply Discourse Relation

(Cause → Effect / Elaboration)



Add Keywords

(focus, break, confident)



Generate 2–3 Sentences



Check Coherence & Tone

b) Three Possible Responses

Response 1:

You can improve your focus by studying in short sessions and taking a small break between them. This will help you concentrate better for the exam, so stay confident that you can do well.

Response 2:

Since your exam is tomorrow, take a short break first and then return with a clear focus. This can help you concentrate better, and you should feel confident about your preparation.

Response 3:

To concentrate better for your exam, remove distractions and focus on one topic at a time. Take a short break when needed, and stay confident in your ability to perform well.

c) Evaluation and Best Response

Response	Coherence	Constraints	Tone	Result
1	High	All	Positive	Good
2	High	All	Positive	Best
3	High	All	Positive	Good

Best: Response 2

Exam Problem

↓

Short Break

↓

Clear Focus

↓

Better Concentration

↓

Confidence

Justification: Response 2 clearly connects the exam → break → focus → concentrate → confident sequence. It satisfies all required keywords, entity coherence, discourse relation, length, consistency, and positive tone.

d) Effect of Violating Two Constraints

If entity coherence is violated:

Using unrelated terms instead of exam, concentrate, and you can make the response less connected to the user's problem.

If response length is violated:

Using more than 3 sentences would fail the required format and make the dialog less concise for the situation.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

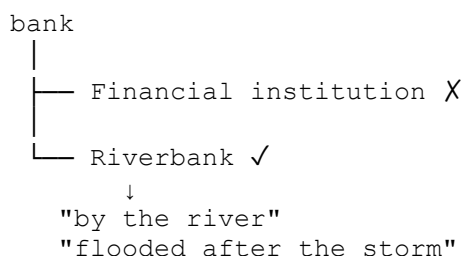
c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence.

a) Word Sense Disambiguation

Ambiguous word: *bank*



Answer: Bank = riverbank.

Justification: The phrases “by the river” and “flooded after the storm” strongly indicate a riverbank, not a financial institution.

b) Predicate Logic Representation

```
river(r)
bank(b)
location(b,r)
storm(s)
flood(b)
quick_action(a)
saved(b,a)
```

Contrast:
flood(b) BUT saved(b,a)

Meaning:

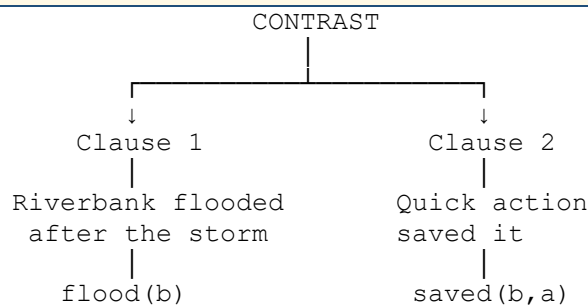
The riverbank is located by the river, it flooded after the storm, and it was saved through quick action.

c) Target Sentence – Simple English

The riverbank flooded after the storm, but quick action saved it.

Key point: “Riverbank” makes the intended meaning of **bank** explicit while preserving the flooding, storm, contrast, and quick action.

d) Discourse Tree



Advantages of Constraint-Based Approach

Context
↓
Constraints
↓
"bank" = Riverbank
↓
Correct Meaning
↓
Coherent Translation

- **Context-aware:** Uses “river” and “flooded” to select the correct sense.
- **Preserves meaning:** Maintains all important entities and relations.
- **Maintains discourse:** Preserves the **contrast** between flooding and being saved.
- **Reduces ambiguity:** Avoids incorrectly translating *bank* as a financial institution.
- **More controllable:** Explicit constraints can enforce logical and semantic consistency, whereas a purely statistical MT system may choose an incorrect sense depending on learned patterns.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

